



GIRRAWEEEN HIGH SCHOOL

2006
YEAR 12 HALF YEARLY
EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work

Total marks – 100

- Attempt Questions 1 – 4
- All questions are NOT of equal value
- Start a separate piece of paper for each question.
- Put you name and the question number at the top of each sheet.

Total marks – 100

Attempt Questions 1 – 4

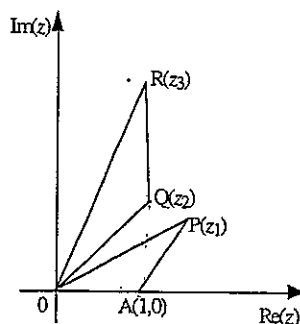
All questions are NOT of equal value

Answer each question on a SEPARATE piece of paper clearly marked Question 1, Question 2, etc. Each piece of paper must show your name.

Question 1 (29 marks) Use a separate piece of paper

Marks

- a) Given that for the complex number z , $|z| = 2$ and $\arg z = \frac{2\pi}{3}$.
- Express z in the form $a + ib$ 2
 - Express $\bar{z}$ in the form $a + ib$ 1
 - Express z^5 in the form $a + ib$ 2
- b) For the complex equation $(5 + 3i)z^2 - (1 - 4i)z + (8 - 2i) = 0$
- Show that the product of the roots is equal to $1 - i$ 3
 - Find the modulus of the product of the roots. 1
 - Find the argument of the product of the roots. 1
- c) Find all the pairs of integers x and y such that $(x + iy)^2 = 5 - 12i$ 4
- d) (i) On an Argand Diagram shade in the region containing all the points representing complex numbers z such that both $|z| \leq 2$ and $\frac{\pi}{4} \leq \arg(z + 2) \leq \frac{\pi}{2}$ 3
- (ii) Find the possible values of $|z|$ and $\arg z$ for such complex numbers z . 3
- e) Interpret geometrically the locus of z given that $\arg\left(\frac{z-2}{z+2}\right) = \frac{\pi}{3}$ 4
- f)



In the diagram above P and Q represent the complex numbers z_1 and z_2 respectively. A is the point $(1,0)$. The triangle OQR is constructed similar to triangle OAP .

Let R represent the complex number z_3

- Show that $|z_3| = |z_1||z_2|$ 2
- Show that $\arg z_3 = \arg z_1 + \arg z_2$ 2
- What does this tell you about the relationship between z_1 , z_2 and z_3 ? 1

Question 2 (29 marks) Use a *separate* piece of paper

Marks

- a) $P(x_1, y_1)$ is a point on the ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$ which has foci at S and S' .
- (i) Determine the eccentricity of the ellipse. 2
 - (ii) Find the coordinates of S and S' and also the equation of the directrices. 2
 - (iii) Prove that $PS + PS' = 4$ 2
 - (iv) Show that the equation of the tangent at P is $\frac{xx_1}{4} + \frac{yy_1}{3} = 1$ 3
 - (v) This tangent meets the nearer directrix at R . If S is the nearer focus to P prove that $\angle PSR = 90^\circ$ 4
- b) $P(x_1, y_1)$ is a point on the hyperbola $9x^2 - 16y^2 = 144$.
- (i) Write down the equation of the tangent at P . 1
 - (ii) Find the coordinates of Q , the point where the tangent cuts the x axis. 1
 - (iii) Show that $\frac{SP}{S'P} = \frac{SQ}{S'Q}$ where S and S' are the foci of the hyperbola. 4
- c) The hyperbola H has the equation $xy = c^2$. $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ are points on the hyperbola H where $p > 0$, $q > 0$ and $c > 0$.
- (i) Prove that the equation of the tangent at P is $x + p^2y = 2cp$ 3
 - (ii) Show that the tangents at P and Q meet at $T\left(\frac{2cpq}{p+q}, \frac{2c}{p+q}\right)$ 3
 - (iii) Suppose further that PQ is perpendicular to OT produced, where O is the origin. Express q in terms of p . 2
 - (iv) Find the locus of T , stating any restrictions that may exist. 2

Question 3 (26 marks) Use a *separate* piece of paper

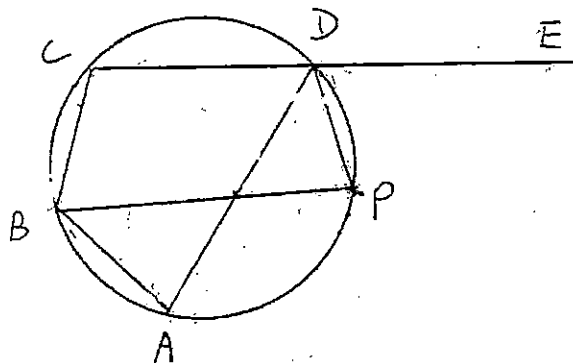
- a) If $2x^4 + 9x^3 + 6x^2 - 20x - 24 = 0$ has a triple root, find all of the roots. 4
- b) Given that α, β and γ are the roots of $x^3 - x^2 + 5x - 3 = 0$, find;
- (i) $\alpha^2 + \beta^2 + \gamma^2$ 3
 - (ii) $\alpha^3 + \beta^3 + \gamma^3$ 3
- c) The equation $x^3 + 3ax + b = 0$ has two equal roots. 4
- Prove that $b^2 + 4a^3 = 0$

Question 3...continued**Marks**

- d) The equation $2x^3 + 3x^2 + x - 5 = 0$ has roots α, β and γ . 4
Find an equation with roots α^2, β^2 and γ^2
- e) (i) Use De Moivre's Theorem to show that $\cos 5\theta = 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta$ 3
(ii) Hence solve $16x^4 - 20x^2 + 5 = 0$ 3
(iii) Hence determine the exact value of $\cos^2 \frac{\pi}{10} \cos^2 \frac{3\pi}{10}$ 2

Question 4 (16 marks) Use a *separate* piece of paper

- a) $ABCD$ is a cyclic quadrilateral. CD is produced to E . P is a point on the circle such that $PA = PC$.



- (i) Prove that PD bisects $\angle ADE$ 4
(ii) If $\angle BAP = 90^\circ$ and $\angle APD = 90^\circ$, explain where the centre of the circle is located. 2
- b) (i) On the same diagram sketch the graphs of $x^2 + y^2 = 1$ and $x^2 - y^2 = 1$, clearly showing the coordinates of any points of intersection with the axes and the equations of any asymptotes. 3
(ii) Shade the region where the inequality $(x^2 + y^2 - 1)(x^2 - y^2 - 1) \leq 0$ 2
- c) In the Pascal triangle, a row consists of the integers $\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \dots, \binom{n}{n}$
Let $p_n = \binom{n}{0} \times \binom{n}{1} \times \binom{n}{2} \times \dots \times \binom{n}{n}$
- (i) Show that $\frac{\binom{n}{r}}{\binom{n-1}{r}} = \frac{n}{n-r}$ 2
- (ii) Prove that $\frac{p_n}{p_{n-1}} = \frac{n^{n-1}}{(n-1)!}$ 3

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log x, \quad x > 0$

Extension 2 Half Yearly 2006 Solutions

Question 1 (29)

$$a) z = 2(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3})$$

$$= 2(-\frac{1}{2} + i \frac{\sqrt{3}}{2})$$

$$= -1 + i\sqrt{3} \quad (2)$$

$$(ii) \bar{z} = -1 - i\sqrt{3} \quad (1)$$

$$(iii) z^5 = 2^5(\cos \frac{10\pi}{3} + i \sin \frac{10\pi}{3})$$

$$= 32(-\frac{1}{2} - i \frac{\sqrt{3}}{2})$$

$$= -16 - 16i\sqrt{3} \quad (2)$$

$$b) (i) \alpha\beta = \frac{8-2i}{5+3i} \times \frac{5-3i}{5-3i}$$

$$= \frac{40-24i-10i-6}{25+9}$$

$$= \frac{34-34i}{34}$$

$$= 1-i \quad (3)$$

$$(ii) |\alpha\beta| = \sqrt{1^2 + (-1)^2}$$

$$= \sqrt{2} \quad (1)$$

$$(iii) \arg(\alpha\beta) = \tan^{-1}(-\frac{1}{1})$$

$$= -\frac{\pi}{4} \quad (1)$$

$$c) (x+iy)^2 = 5-12i$$

$$x^2 + 2ixy - y^2 = 5-12i$$

$$x^2 - y^2 = 5 \quad 2xy = -12$$

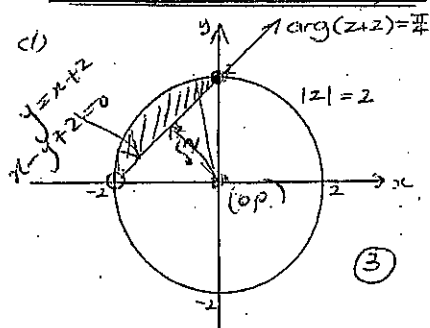
$$x^2 - (-\frac{6}{x})^2 = 5 \quad y = -\frac{6}{x}$$

$$x^4 - 5x^2 - 36 = 0$$

$$(x^2-9)(x^2+4) = 0$$

$$x = \pm 3 \text{ or no real solutions}$$

$$x = 3, y = -2 \text{ or } x = -3, y = 2 \quad (4)$$



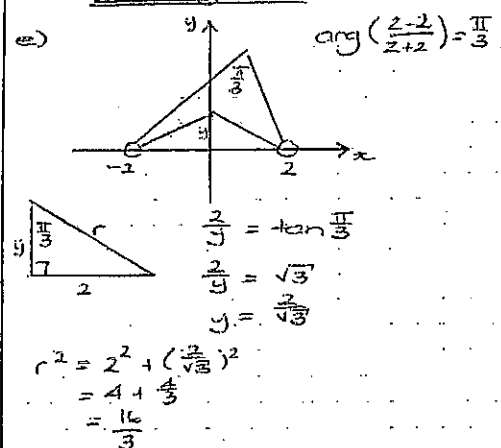
$$(ii) \min |z| = \frac{|0+0+2|}{\sqrt{1^2+1^2}}$$

$$= \frac{2}{\sqrt{2}}$$

$$= \sqrt{2}$$

$$\therefore \sqrt{2} \leq |z| \leq 2 \quad (2)$$

$$\frac{\pi}{2} \leq \arg z < \pi \quad (3)$$



locus is major arc of circle

$$x^2 + (y - \frac{2}{\sqrt{3}})^2 = \frac{16}{3} \quad (4)$$

cut off by chord joining $(\pm 2, 0)$, excluding these points.

f) $\frac{OQ}{OA} = \frac{OR}{OP}$ (corresponding sides in $\triangle OAB$)

$$\frac{|z_1|}{1} = \frac{|z_3|}{|z_2|} \quad \text{in } \triangle OAB$$

$$|z_3| = |z_1||z_2| \quad (2)$$

(ii) $\angle POR = \angle POA$ (corresponding $\angle$ s in $\triangle OAB$)

$\angle AOR \neq \angle AOR + \angle POR$ (common $\angle$ s)

$\therefore \angle AOR = \angle AOR + \angle POA$

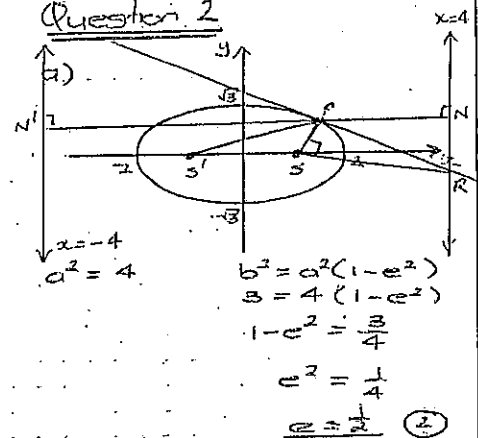
$$\arg z_3 = \arg z_1 + \arg z_2 \quad (2)$$

(iii) $z_3 = z_1 \times z_2 \quad (1)$

$$y = x+2$$

$$x-y+z=0$$

Question 2



(ii) foci $(\pm 2, 0)$

directrices: $x = \pm 4$ (2)

(iii) $PS + PS' = e(PN + PN')$

$$= \frac{1}{2}(8)$$

$$= 4 \quad (2)$$

(iv) $\frac{x^2}{4} + \frac{y^2}{3} = 1$

$$\frac{x}{2} + \frac{2y}{3} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{3x}{4y}$$

at P, $\frac{dy}{dx} = -\frac{3x_1}{4y_1}$

$$y - y_1 = -\frac{3x_1}{4y_1}(x - x_1)$$

$$4y_1 - 4y_1^2 = -3xx_1 + 3x_1^2$$

$$3xx_1 + 4y_1^2 = 3x_1^2 + 4y_1^2$$

$$\frac{xx_1}{4} + \frac{yy_1}{3} = \frac{x_1^2}{4} + \frac{y_1^2}{3}$$

$$\frac{xx_1}{4} + \frac{yy_1}{3} = 1 \quad (3)$$

(v) when $x = 4$, $x_1 + \frac{4y_1}{3} = 1$

$$\frac{4y_1}{3} = 1 - x_1$$

$$y = \frac{3-3x_1}{4}$$

$$R(4, \frac{3-3x_1}{4})$$

$$m_{PS} = \frac{y_1}{x_1 - 1}$$

$$m_{RS} = \frac{y_1}{x_1 - 1}$$

$$= \frac{1-x_1}{4}$$

$$m_{PS} \times m_{RS} = \frac{y_1}{x_1 - 1} \times \frac{1-x_1}{4}$$

$$= -1$$

$\therefore PS \perp RS$

$\angle PSR = 90^\circ \quad (4)$

b) $9x_1x - 16y_1y = 144 \quad (1)$

(ii) $y = 0$, $x = \frac{144}{9x_1}$

$$= \frac{16}{x_1}$$

$Q(\frac{16}{x_1}, 0) \quad (1)$

(iii) $SP = ePN$

$$= e(x_1 - \frac{4}{e})$$

$$S'P = ePN'$$

$$= e(x_1 + \frac{4}{e})$$

$$\frac{SP}{S'P} = \frac{x_1 - \frac{4}{e}}{x_1 + \frac{4}{e}}$$

$$= \frac{ex_1 - 4}{ex_1 + 4}$$

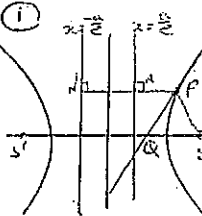
$$SQ = 4e - \frac{16}{x_1} \quad S'Q = 4e + \frac{16}{x_1}$$

$$\frac{SQ}{S'Q} = \frac{4e - \frac{16}{x_1}}{4e + \frac{16}{x_1}}$$

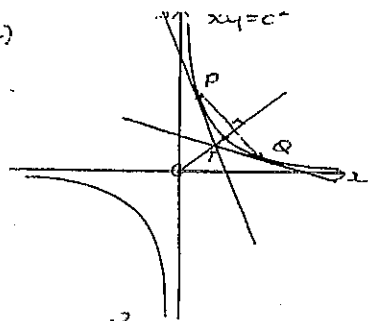
$$= \frac{4ex_1 - 16}{4ex_1 + 16}$$

$$= \frac{ex_1 - 4}{ex_1 + 4}$$

$$= \frac{SP}{S'P} \quad (4)$$



c)



$$(i) y = \frac{c}{x}$$

$$y' = -\frac{c}{x^2}$$

$$\text{at } P, y' = -\frac{c}{x^2}$$

$$y - \frac{c}{x} = -\frac{1}{x^2}(x - cp)$$

$$p^2 y - cp = -x + cp$$

$$x + p^2 y = 2cp \quad (3)$$

$$(ii) x + p^2 y = 2cp \quad (-)$$

$$\frac{x + q^2 y}{(p^2 - q^2)y} = \frac{2cp}{2c(p-q)}$$

$$\frac{y}{p+q} = \frac{2cp}{p+q}$$

$$x + \frac{2cp^2}{p+q} = 2cp$$

$$x = \frac{2cp^2 + 2cpq - 2cp}{p+q}$$

$$= \frac{2cpq}{p+q}$$

$$T \left\{ \frac{2cpq}{p+q}, \frac{2c}{p+q} \right\} \quad (3)$$

$$(iii) m_{PQ} \times m_{OT} = -1$$

$$\frac{\frac{c}{p} - \frac{c}{q}}{cp - cq} \times \frac{\frac{2c}{p+q}}{\frac{2cpq}{p+q}} = -1$$

$$\frac{q-p}{p-q} \times \frac{1}{pq} = -1$$

$$\frac{1}{p^2 q^2} = -1$$

$$p^2 q^2 = 1$$

$$pq = 1 \quad (p, q > 0)$$

$$q = \frac{1}{p} \quad (2)$$

$$(iv) x = pqy$$

$$x = y$$

$$\text{locus is } y = xc, \quad 0 < x \leq c \quad (2)$$

Question 3 (26)

$$a) P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24$$

$$P'(x) = 8x^3 + 27x^2 + 12x - 20$$

$$P''(x) = 24x^2 + 54x + 12$$

$$= 6(4x^2 + 9x + 2)$$

$$= 6(4x+3)(x+2)$$

$$\text{triple root } x = -\frac{3}{4} \text{ or } x = -2$$

$$P'(-2) = 0, \quad P'(-\frac{3}{4}) = 0$$

$$\therefore x = -2 \text{ is triple root!}$$

$$2x^4 + 9x^3 + 6x^2 - 20x - 24 = 0$$

$$(x+2)^3(2x-3) = 0$$

$$\therefore \text{roots are } -2, -2, -2, \frac{3}{2} \quad (4)$$

$$b) x^3 - x^2 + 5x - 3 = 0$$

$$(i) \Delta x^2 = (\Delta x)^2 - 2\Delta x \Delta y$$

$$= (1)^2 - 2(5)$$

$$= -9 \quad (3)$$

$$(ii) \Delta x^3 - \Delta x^2 + 5\Delta x - 9 = 0$$

$$\Delta x^3 = \Delta x^2 - 5\Delta x + 9$$

$$= (-9) - 5(1) + 9$$

$$= -5 \quad (3)$$

$$c) x^3 + 3ax + b = 0$$

$$\text{Let roots be } x, \alpha, \beta$$

$$2x + \beta = 0 \quad \alpha^2 \beta = -b$$

$$\beta = -2x \quad -2\alpha^3 = -b$$

$$\alpha^3 = \frac{b}{2}$$

$$\alpha = \sqrt[3]{\frac{b}{2}}$$

$$\alpha^2 + \alpha\beta + \alpha\beta = 3a$$

$$\alpha^2 + 2\alpha^2 - 2\alpha^2 = 3a$$

$$-3\alpha^2 = 3a$$

$$\alpha^2 = -a$$

$$\left(\sqrt[3]{\frac{b}{2}}\right)^2 = -a$$

$$\frac{b^2}{4} = -a^3$$

$$b^2 + 4a^3 = 0 \quad (4)$$

$$d) 2x^3 + 3x^2 + x - 5 = 0$$

$$\text{let } y = x^{\frac{1}{2}}$$

$$x = y^2$$

$$2y^{\frac{3}{2}} + 3y + y^{\frac{1}{2}} - 5 = 0$$

$$2y^{\frac{3}{2}} + y^{\frac{1}{2}} = 5 - 3y$$

$$y(2y+1)^{\frac{1}{2}} = (5-3y)^{\frac{1}{2}}$$

$$4y^3 + 4y^2 + y = 25 - 30y + 9y^2$$

$$4y^3 - 5y^2 + 31y - 25 = 0 \quad (4)$$

$$e) \cos 5\theta = (c + is)^5$$

$$= c^5 + 5ic^4s - 10c^3s^2 - 10ic^2s^3$$

$$+ 5cs^4 + is^5$$

$$\text{equating reals}$$

$$\cos 5\theta$$

$$= \cos^5 \theta - 10\cos^3 \theta \sin^2 \theta$$

$$+ 5\cos \theta \sin^4 \theta$$

$$= \cos^5 \theta - 10\cos^3 \theta (1 - \cos^2 \theta)$$

$$+ 5\cos \theta (1 - 2\cos^2 \theta + \cos^4 \theta)$$

$$= \cos^5 \theta - 10\cos^3 \theta + 10\cos^5 \theta$$

$$+ 5\cos \theta - 10\cos^3 \theta + 5\cos^5 \theta$$

$$= 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta \quad (3)$$

$$(ii) 16x^4 - 20x^2 + 5 = 0$$

$$\text{let } x = \cos \theta$$

$$16\cos^4 \theta - 20\cos^2 \theta + 5 = 0$$

$$\cos 5\theta = 0$$

$$5\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \frac{9\pi}{2}$$

$$\theta = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{5\pi}{10}, \frac{7\pi}{10}, \frac{9\pi}{10}$$

$$\theta = \frac{\pi}{10}, \frac{3\pi}{10}, \frac{5\pi}{10}, \frac{7\pi}{10}, \frac{9\pi}{10} \quad (3)$$

$$x = \cos \frac{\pi}{10}, \cos \frac{3\pi}{10}, \cos \frac{5\pi}{10}, \cos \frac{7\pi}{10}, \cos \frac{9\pi}{10}$$

NOTE: $\cos \frac{\pi}{2} = 0$ not a solution of original equation

$$(iii) \cos \frac{\pi}{10} \cos \frac{3\pi}{10} \cos \frac{5\pi}{10} \cos \frac{7\pi}{10} \cos \frac{9\pi}{10} = \frac{1}{16} \quad (2)$$

$$\cos \frac{\pi}{10} = -\cos \frac{9\pi}{10}$$

$$\cos \frac{3\pi}{10} = -\cos \frac{7\pi}{10}$$

$$\therefore \cos \frac{2\pi}{10} \cos \frac{4\pi}{10} = \frac{1}{16} \quad (2)$$

Question 4 (16)

$$a) PA = PC \quad (\text{given})$$

$$\therefore \angle CBP = \angle ABP \quad (\text{chords subtend } \angle \text{'s at circumference})$$

$$\angle CBP = \angle PDG \quad (\text{exterior } \angle \text{ cyclic quadrilateral})$$

$$\angle ABP = \angle ADP \quad (\angle \text{'s in same segment})$$

$$\therefore \angle PDG = \angle ADP$$

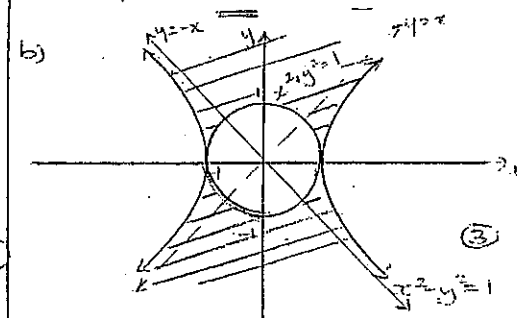
$$\text{ie PD bisects } \angle ADE \quad (4)$$

$$(ii) BP \text{ is a diameter } (\angle BAP = 90^\circ)$$

$$\text{AD is a diameter } (\angle APD = 90^\circ)$$

$$\therefore \text{centre is intersection of BP and AD}$$

$$\text{BP and AD}$$



$$(ii) \text{ region would be when}$$

$$x^2 + y^2 \leq 1 \text{ and } x^2 - y^2 \geq 1 \quad \text{not possible}$$

$$\text{or}$$

$$x^2 + y^2 \geq 1 \text{ and } x^2 - y^2 \leq 1 \quad (2)$$

$$c) \frac{\binom{n}{r}}{\binom{n-1}{r}} = \frac{n!}{r!(n-r)!} \times \frac{r!(n-r-1)!}{(n-1)!}$$

$$= \frac{n}{n-r} \quad (2)$$

$$(ii) \frac{P_n}{P_{n-1}} = \frac{\binom{n}{0} \binom{n}{1} \binom{n}{2} \dots \binom{n}{n-1} \binom{n}{n}}{\binom{n-1}{0} \binom{n-1}{1} \binom{n-1}{2} \dots \binom{n-1}{n-1} \binom{n-1}{n}}$$

$$= \frac{\binom{n}{1}}{\binom{n-1}{1}} \times \frac{\binom{n}{2}}{\binom{n-1}{2}} \times \dots \times \frac{\binom{n}{n-1}}{\binom{n-1}{n-1}}$$

$$= \frac{n}{n-1} \times \frac{n}{n-2} \times \dots \times \frac{n}{n-(n-1)}$$

$$= \frac{n}{n-1} \times \frac{n}{n-2} \times \dots \times \frac{n}{1}$$

$$= \frac{n^{n-1}}{(n-1)!} \quad (3)$$